Response to Reviewers

Learning Annotation-Driven Latent Structure for Multi-Modal Fusion

We thank the reviewers for their careful reading and constructive suggestions. We have revised the manuscript accordingly; all new content is marked in red in the revised PDF for ease of review. Below we respond to each comment in turn and indicate the specific additions.

Reviewer 1

Comment 1. *Clarify the essential difference between MMCTL and supervised contrastive learning or cross-modal triplet loss, especially the added contribution of continuous similarity constraints.*

Response. We rewrote the Related Work section and added an explicit statement of the contribution. Unlike binary supervised contrastive learning and fixed-margin cross-modal triplets, MMCTL orders embedding distances by graded, continuous annotation proximity rather than binary co-membership. Concretely, MMCTL adds:

- a joint annotation vector with feature-wise normalization and per-task weights (Eqs. 1-3);
- a margin-free continuous triplet whose separation target is the annotation-distance gap (Eq. 7), with no tuned margin;
- a pairwise distance-alignment regularizer matching embedding to annotation distances (Eq. 8);
- extension to a multi-modal shared space where same-object, different-modality pairs have zero annotation distance, directly pulling cross-modal embeddings together.

Comment 2. *The paper claims MMCTL learns a more interpretable latent space; visualizations or cross-modal retrieval examples are recommended.*

Response. We added embedding-space visualizations and quantitative interpretability metrics:

- Figure 1: AWIR fused embeddings (PCA and t-SNE) across all fusion methods, colored by class and sized by bounding-box area;
- Figure 3: per-class "zoom" panels colored by the continuous label, showing smooth within-class transitions;
- Table 1: standard metrics -- silhouette (class separation) and Spearman distance-preservation (continuous attributes).

Comment 3. *Specify the rules for selecting positive and negative samples in continuous triplets, as well as the similarity thresholds or margin settings.*

Response. Clarified in the Shared Latent Space subsection. We use a batch-all strategy: every in-batch index is an anchor, and a pair (p,q) is a valid (positive, negative) pair iff p is closer to the anchor than q in joint annotation space (Eq. for triplet ordering). Positives/negatives are defined solely by annotation-distance ordering; there is no hard similarity threshold and no tuned margin (the loss is margin-free).

Comment 4. *How are annotation attributes such as category, object size, and height normalized and weighted? Does the weighting strategy affect the final results?*

Response. The Problem Setup gives feature-wise min-max normalization, task-length rescaling, and fixed per-task weights (Eqs. 1-3; AWIR w_class=0.5, NEON w_class=1.0). We added a class-weight sensitivity study:

- Table 3 and Figure 4 sweep w_class in {0.5, 0.8, 1.0, 1.5};

- accuracy rises monotonically with w_class (74.9% -> 84.6%) while box/height regression stays stable, confirming the weight acts as an interpretable semantic-vs-geometric trade-off.

Comment 5. *What is the rationale for freezing the encoder in the two-stage training scheme? A comparison with end-to-end fine-tuning would help validate this.*

Response. We justify freezing as a fair fusion-evaluation protocol (it isolates fusion geometry from unequal encoder capacity) and added a controlled comparison on NEON from the same pretrained encoders:

- frozen: 81.99 +/- 1.02% accuracy; end-to-end: 80.66 +/- 1.35%;
- end-to-end yields only marginal gains on the continuous heads, so freezing is justified and substantially cheaper.

Comment 6. *The baseline set is limited. Representative multimodal methods such as supervised contrastive learning, cross-attention, or Transformer-based fusion should be included.*

Response. We removed the DeepSet baselines and added three stronger, representative fusion baselines with citations: supervised contrastive fusion (SupCon), cross-attention pooling, and a Transformer encoder over modality tokens. They appear in Tables 1 and 2 and in the embedding visualizations.

Comment 7. *Report standard deviations and statistical significance tests to verify whether MMCTL's improvements are stable and reliable.*

Response. All results tables now report mean +/- std over the cross-validation folds, and we added paired one-sided Wilcoxon signed-rank tests:

- MMCTL's continuous-geometry advantage is significant on AWIR position (p=0.0078), with a positive trend on box (p=0.098);
- we are explicit that MMCTL's contribution is balanced coherence across metrics rather than winning every individual metric.

Reviewer 2

Comment 1. *State where the substantive increment relative to prior work lies -- the joint annotation vector/weighting (Eqs. 1-3), the margin-free continuous triplet (Eq. 7), the distance-alignment regularizer (Eq. 8), or the extension from single-modal to multi-modal shared latent space.*

Response. The primary increment is the extension to a multi-modal shared latent space in which same-object, different-modality pairs share an identical joint annotation vector (zero annotation distance), directly coupling cross-modal embeddings. This is accompanied by the joint annotation vector construction/weighting (Eqs. 1-3), the margin-free continuous triplet (Eq. 7), and the distance-alignment regularizer (Eq. 8). Related Work now states this explicitly relative to our prior unimodal work.

Comment 2. *The introduction uses gradient conflict and task dominance as motivation but provides no supporting experiments; revise the introduction to highlight the problem the paper solves.*

Response. We revised the Introduction to remove the unsupported gradient-conflict/task-dominance motivation and instead frame the problem directly: task-driven fusion shapes latent geometry only indirectly, so embedding distances are not constrained to reflect annotation-defined similarity. The motivation now matches the problem addressed and the evidence provided.

Comment 3. *Discuss the differences between MMCTL and existing soft/continuous similarity metric learning and adaptive-margin triplet losses.*

Response. Related Work now contrasts MMCTL with continuous/soft-similarity metric learning (Kim et al.) and adaptive-margin triplets (Ha et al.): MMCTL is margin-free (its separation target is the annotation-distance gap, with no tuned margin) and operates in a multi-modal shared space, which those methods do not.

Comment 4. *MMCTL also uses task labels; the difference from baselines is the loss form, not the information used. Add cross-dataset/cross-region transfer or missing-modality experiments.*

Response. We agree the distinction is the inductive bias (loss form), not the label information, and we now state this explicitly. We added missing-modality experiments:

- zeroing one modality half of the MMCTL joint embedding and re-probing yields graceful degradation (drop-thermal 84.7%, drop-RGB 86.2%) that stays above the unimodal baselines (~78-80%), evidencing complementary but non-redundant modality use;
- cross-region transfer is noted as future work in the Conclusion.

Comment 5. *The batch-all strategy is used but the number of triplets and computational cost are not reported. Report total triplets, training time per epoch, and GPU memory usage.*

Response. We added a triplet-compute analysis. With batch size 32 and modality concatenation (64 embeddings), the batch-all construction yields at most 249,984 ordered triplets per batch (~2.0M per epoch) before the annotation-order mask; a microbenchmark measures ~1.7 s per training step (batch 8) on an Apple M4 CPU. Our runs were CPU-based, so peak GPU memory was not measured; we state this explicitly and will add GPU timing/memory if GPU logs are required.

Comment 6. *Clarify the relationship between "k-fold cross-validation within the training set" and "averaged over 5 runs" in the caption of Figure 3.*

Response. Clarified in the Experimental Setup: the "runs" reported in the captions ARE the cross-validation folds (8 for AWIR, 5 for NEON), and each mean +/- std is computed across these folds. The ambiguous "runs" wording has been reconciled with the k-fold description.

Comment 7. *Some downstream results need more detailed discussion: add error analysis of typical failure cases and explain when the method is effective and when it is not.*

Response. We expanded the downstream discussion to explain when MMCTL helps and when it does not:

- class-supervised baselines win raw class separation, and a directly sensing modality (LiDAR) wins its own continuous attribute, but each fails the other axis; MMCTL is the only method balanced across both;
- the per-class within-class analysis (Figure 3) identifies classes with strong continuous transitions (e.g., PIPA2 canopy height) versus weaker ones, providing a class-level effectiveness/error analysis.

Comment 8. *Update references to their formally published versions, e.g., reference [8].*

Response. We are updating the bibliography to cite formally published versions in place of arXiv preprints where available, including reference [8]. (This audit is being finalized against the compiled reference list.)